DS 574 Algorithmic Mechanism Design **Voting Worksheet**
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The Difficulty of Social Choice

We want to choose one outcome from individual rankings, without payments. What should a good rule respect: first choices, the entire ranking, head-to-head majorities, or incentives to report truthfully? These goals lead to different rules, even when everyone reports honestly.

Definition 1. There is a set A of m *alternatives* (candidates) and a set of n voters. Each voter i has a *preference* $\succ_i$, a strict total order on A, and the tuple $\succ = (\succ_1, \ldots, \succ_n)$ is a *preference profile*. A *voting rule* is a function f from profiles to a single winner in A; a *social welfare function* maps every profile to a full ranking of A.

Fix tie-breaking in advance; in the examples below, b wins a tie with a. A rule's output depends on both the ballots and the rule used to combine them. Which two goals would we most want an election rule to satisfy?

Plurality: First Choices and Incentives

Plurality. Each voter names one alternative, their first choice; the alternative named most often wins. A majority is not required.

	5 voters	4 voters	2 voters
1st choice	a	c	b
2nd choice	b	d	c
3rd choice	c	b	a
4th choice	d	a	d

Who wins? Does that candidate have a majority of first choices?

U.S. example. Maine elects its governor by plurality in the general election, even though it uses RCV in federal general elections.

Can One Voter Benefit by Misreporting?

A rule is *strategyproof* if reporting truthfully is a dominant strategy: fixing all other reports, no voter can obtain a strictly preferred outcome by changing their report. *Historical anchor: Florida, 2000.* Bush's certified lead over Gore was 537 votes; Nader received 97,488. This motivates concern about vote splitting, but these single-choice ballots do not identify voters' full rankings or establish who would win under another rule.

	4 voters	3 voters	2 voters
1st choice	a	b	c
2nd choice	b	c	b
3rd choice	c	a	a

Truthful counts $(4, 3, 2)$ elect a. Can one voter obtain an outcome they prefer by misreporting? Keep the other reports and the tie-break fixed.

With two alternatives, why is majority rule strategyproof?

Borda: Use the Whole Ranking

Borda count. With m alternatives, each ballot awards $m - 1$ points to its first choice, $m - 2$ to its second, down to 0 for last place. The highest total wins. Return to the same eleven voters:

	5 voters	4 voters	2 voters
1st choice	a	c	b
2nd choice	b	d	c
3rd choice	c	b	a
4th choice	d	a	d

The Borda totals for (a, b, c, d) are $(17, 20, 21, 8)$, so c wins. Why does this rule choose differently from plurality? No new tally is needed.

American ranked-points example. Heisman Trophy ballots award three, two, and one points to a voter's top three players. This is a Borda-style scoring rule, not the complete-ranking Borda rule analyzed here.

Condorcet: Head-to-Head Majorities

A *Condorcet winner* beats every other alternative in a head-to-head majority comparison. The Condorcet criterion asks a voting rule to select that alternative whenever one exists. Return to the same eleven voters:

	5 voters	4 voters	2 voters
1st choice	a	c	b
2nd choice	b	d	c
3rd choice	c	b	a
4th choice	d	a	d

Candidate b beats a by 6–5 and beats c and d by 7–4. Compare the plurality, Borda, and Condorcet winners. Which notion of support does each use?

What If Majority Comparisons Cycle?

	Voter 1	Voter 2	Voter 3
1st choice	a	b	c
2nd choice	b	c	a
3rd choice	c	a	b

Each contest is 2–1: a beats b, b beats c, and c beats a. What prevents the head-to-head criterion from choosing a winner? What must a complete voting rule do here?

Instant Runoff: Elimination and Transfers

Instant-runoff voting (IRV), called RCV here. If a candidate has a strict majority of the active first choices, elect them. Otherwise eliminate the candidate with the fewest first choices and transfer each of their ballots to its highest-ranked remaining candidate. Repeat, with tie-breaking fixed beforehand. Our example uses complete rankings. *U.S. examples.* Maine uses RCV for federal general elections and state primaries, but not general elections for governor or the state legislature. Alaska pairs a top-four primary with RCV general elections (first used in 2022). NYC uses RCV for municipal primaries and special elections, first used in 2021, rather than its general elections.

A different route to a majority. In Georgia U.S. Senate elections, there is a separate top-two runoff if no candidate wins the required majority. This is a second election, not instant elimination and transfers from one ranked ballot.

	6 voters	5 voters	4 voters	2 voters
1st choice	a	c	b	b
2nd choice	b	a	c	a
3rd choice	c	b	a	c

Supplied tally, in order (a, b, c): $(6, 6, 5)$. Eliminate c; its five ballots transfer to a, who wins 11–6.

Now raise a above b on the two-voter bloc's ballots. The new first counts are $(8, 4, 5)$. Eliminate b; its four ballots transfer to c, who wins 9–8.

Explain why the increased support for a changes the elimination order and makes a lose. Does this seem like a desirable response to raising a candidate?

A rule is *monotone* if raising its winner on ballots, with all other comparisons fixed, cannot make that candidate lose. Which step above witnesses the failure?

A General Obstacle

Theorem 1 (Gibbard '73-Satterthwaite '75)**.** *With at least three alternatives, every deterministic, onto, strategyproof voting rule on all strict preference profiles is a dictatorship.*

Onto: every alternative wins at some profile. A dictatorship always chooses one fixed voter's top alternative. The theorem assumes all strict profiles are permitted.

Which assumption lets two-alternative majority rule avoid this theorem? What could we restrict while keeping at least three alternatives?

Single-Peaked Preferences and the Median Rule

Suppose alternatives have a common fixed order. Each voter has a favorite, their *peak*, and prefers alternatives nearer their peak on each side to those farther away on that same side. Comparisons across the two sides need not be determined by distance.

With at least three voters and an odd total, let each report a peak and select the median reported peak. Fix the other four reports at $2, 4, 6, 8$ on $A = \{1, \ldots, 9\}$.

Mark the outcomes one voter can cause by changing their report. Why does reporting their true peak select the best attainable outcome, whether that peak lies inside, below, or above the attainable interval?

What restriction restores strategyproofness? What difficulty from the earlier examples remains a reason to be careful about the model?